

Resistance Is Futile: The Rise of the Battery Electric Vehicle

Tesla's meteoric rise has catapulted the company to stardom. In less than 20 years, the electric car maker went from an upstart, influenced by GM's EV1 program, to the most valuable car company in America. It's also one of the more polarizing companies out there—it seems most people either love Tesla or hate it. For those who love Tesla, it can do no wrong. Elon Musk is an almost messianic figure who will single-handedly deliver us from the evils of a dirty transportation future. Of course, his flamethrowers and rockets (the latter developed by one of his other companies, SpaceX) are predicated on combustion, but who's counting? And similar to the Prius in the late 1990s, driving a Tesla is as much of a fashion statement as anything else. Those who drive them are willing to put up with the inconveniences of a BEV, simply because owning one says, "I care about the environment." Not to mention that big screen!

I [Kelly] have witnessed this firsthand. A couple of years ago, I was set to deliver a keynote address on the future of the internal combustion engine in San Diego. The night before my speech, I hitched an Uber and was lucky enough to be picked up by a Tesla Model 3! "I'm going to apologize ahead of time," I told the driver. "I'm kind of into cars and I have a lot of questions for you." He was a good sport and was very excited to talk about his car. I proceeded to ask him a number of questions: "How is the range? How is the charge time? What's up with these door handles?" Each question was answered with a high level of frustration, but it didn't matter. He loved that big screen. He loved his cool car. He loved Tesla.

To be fair, there is a lot to love about Tesla. They have single handedly pushed the BEV forward, both in terms of technology and cool factor, to a state of cultural reverence it hasn't experienced in more than 100 years. But while these sexy vehicles

(literally, as the model names are S, 3, X, Y) are marketed¹ as planet-saving, what they seem to really be are high-acceleration cars for rich people. And as we will see throughout the rest of this book, BEVs are no panacea—they have their own share of negative environmental and health effects.

Whether you love them or hate them, BEVs are part of the transportation landscape, both now and in the future. It may seem as if they burst onto the scene in just the last decade or so, but as we learned in Chapter 2, they've been around for a very long time. We will review the bumpy journey of the BEV later in this chapter, but first we need to understand how they differ from today's conventional vehicles.

Electric Vehicles 101

What *is* an electric vehicle? That's easy—a car “powered by” electricity! But while that sounds simple, pinning down precisely what it means can be a bit of a challenge. This is mainly due to the confusion around the words “electric” and “electrified.” These two words, separated by just a few letters, are often (understandably) assumed to be synonymous. In reality, however, they are very different. Here's what we mean: an electric car is electrified, but an electrified car isn't always electric (try saying that ten times fast!). Make sense? Let's examine this a bit further.

Electric versus Electrified

An *electric* powertrain is commonly understood to mean a BEV, which runs solely on an electric motor. (Strictly, electric powertrains also include fuel-cell electric vehicles [FCEVs], but we'll discuss those in Chapter 8.) In contrast, an *electrified* powertrain includes BEVs, but also cars that have an ICE and some degree of electrification, such as mild hybrid electric vehicles (MHEVs), full hybrid electric vehicles (FHEVs), and plug-in hybrid electric vehicles (PHEVs). We'll dig deeper into the differences between various types of hybrid vehicles later in this chapter, but [Table 6.1](#) provides a brief overview. By the way, here's a quick way to tell if a car is fully electric or not. Look at the back of the car. Does it have a tailpipe? If the answer is yes, then it's not a BEV.

The confusion between electric and electrified vehicles became apparent as car manufacturers began to announce they would electrify their fleets. For example, in July 2017 (a.k.a., “the year combustion died”), the press caught on to Volvo's future powertrain plans. Headlines such as “Volvo to make only electric cars from 2019, marking the end of the petrol engine” and “Historic end for combustion: Volvo says all of its cars will use electric after 2019” were everywhere. Even the BBC reported that Volvo was “the first traditional carmaker to signal the end of the internal combustion engine” [2]. These headlines paint an ICE-less future for Volvo, and in the media's defense, Volvo's own press release was titled “Volvo cars to go all electric” [3]. However, it only requires reading the first sentence of that release to see that from 2019, all of

¹ From what we've read, Tesla actually has no marketing department! One report we found [1] analyzed the social media ad spending of various auto companies and found that, in contrast to other car companies who use paid social media campaigns, Tesla's social strategy is strictly organic.

TABLE 6.1 Overview of the various vehicle types discussed in this book.

Type	Acronym	Features	Includes an ICE?
Internal combustion engine vehicle	ICEV	Full internal combustion engine driving	Yes
Micro hybrid electric vehicle	μHEV	Stop/start technology	Yes
Mild hybrid electric vehicle	MHEV	μHEV + regenerative braking for energy recovery and electric torque assistance (assist acceleration from stop)	Yes
Full hybrid electric vehicle	FHEV	MHEV + electric driving with battery charged from the ICE	Yes
Plug-in hybrid electric vehicle	PHEV	FHEV + electric driving with battery charged from the grid	Yes
Extended range electric vehicle	EREV	PHEV, but the ICE is only used as a generator to recharge the battery	Yes
Battery electric vehicle	BEV	Full electric driving with battery charged from the grid	No
Fuel-cell electric vehicle	FCEV	Full electric driving with hydrogen fuel cell producing electricity	No

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Volvo’s vehicles were to be fitted with an electric motor, not that all cars would be fully electric. That can include PHEVs, FHEVs, and MHEVs in addition to BEVs. In other words, ICE vehicles with even just a small amount of electrification were part of their future! Indeed, Volvo stated that all new cars from 2019 would be electrified and not electric. It is important to note that at the time of writing (in 2020), conventional ICEVs are still available to purchase from Volvo. This speaks to the resiliency of the internal combustion engine and the fallibility of marketing.

Similarly confusing headlines about other automakers soon followed. When the media stated that Toyota would “electrify their entire vehicle lineup by 2025” [4], it only took a quick visit to the Toyota Newsroom website to learn what this really meant. It did not mean that all Toyota vehicles would be battery electric by 2025. It didn’t even mean that all vehicles would be electrified in 2025. It meant that all models would have an electrified *option* by 2025. Toyota’s press release went on to say that by around 2030, they aimed to have sales of more than 5.5 million electrified vehicles, including more than 1 million BEVs and FCEVs. If they meet this goal, a quick estimate reveals that around 8% of their sales will be BEVs and FCEVs and about 44% will be hybrids. This leaves 48% as conventional vehicles. That means that 92% of Toyota’s fleet will still be equipped with internal combustion engines!

Around the same time, BMW announced they had reached their goal of selling 100,000 electric cars globally [5]. The company celebrated by lighting up their headquarters in Munich to look like four giant batteries (a bit ironically, the building is known as the “four-cylinder,” in reference to a four-cylinder internal combustion engine). This was undoubtedly a significant milestone, but again, it’s important to put that sales number into context. When BMW said “electric” cars, they meant both

BEVs and PHEVs. Of the 103,080 electric cars BMW sold in 2017, only about 30% were BEVs. Thus, 70% of the electrified cars still contained an ICE. Additionally, BMW’s electric vehicle sales in 2017 accounted for just 4% of their total vehicle sales, and BEV sales on their own accounted for just over 1% [6]. Despite the increasing popularity of electric vehicles, conventional ICE vehicles still make up the vast majority of new car sales today.

Degrees of Electrification

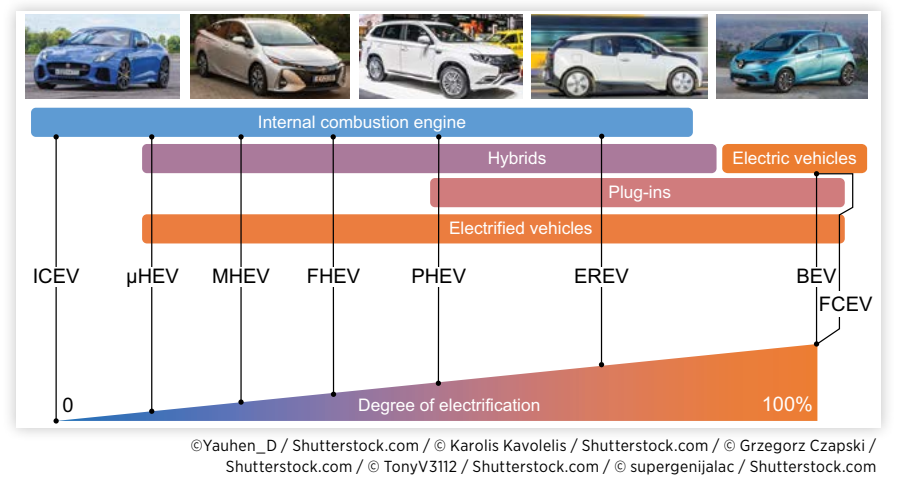
To try to avoid confusion in this chapter, we have included a table that defines the many acronyms for electrified vehicles (Table 6.1) and a graphic depicting the approximate degrees of electrification for each vehicle type, going from 0% to 100% (Figure 6.1).

Let’s drive up the electrification ramp in Figure 6.1. We start with an ICEV, also known as a “conventional vehicle.” We briefly described how internal combustion engines worked in Chapter 2 and will spend more time with them later on. A majority of ICEVs use gasoline for fuel, but there are also diesel vehicles and alternative fuel vehicles that run on CNG, liquefied petroleum gas (LPG), etc. Of course, ICEVs are not battery-free. Their lead-acid batteries help start the engine and power the car’s electrical systems. These auxiliary batteries are also found in electric and hybrid vehicles for powering 12 V accessories, while much larger batteries power the electric motors.

The next step in Figure 6.1 is still a standard ICE-powered vehicle, but it also includes stop/start technology. This is sometimes marketed as a micro hybrid vehicle (μHEV). Micro hybrids include a robust battery and an integrated starter/generator used to shut down and restart the engine. With this type of car, your engine stops when idle and starts again when you release the brake pedal. This technology can reduce fuel consumption by 3–10%, with the largest reductions during city driving [7].

MHEVs are similar to micro hybrids except they store energy recovered during braking and use it for acceleration assistance. They incorporate a 48 V battery and

FIGURE 6.1 Degrees of electrification of the vehicles discussed in this book.



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an electric motor/generator that runs in a parallel hybrid configuration with the ICE. MHEVs can provide fuel efficiency increases of up to 20% compared to ICEVs [8].

In FHEVs, electric driving is possible. The battery is charged from the internal combustion engine. As such, they are sometimes referred to as “self-charging” vehicles since there is no need for an external power source. The ICE and the electric motor run in a parallel hybrid configuration, meaning that both devices can drive the wheels directly. Fuel efficiency increases of 25–35% have been achieved with full hybrids [9]. The most efficient full hybrids can reach 50–60 mpg combined city/highway [10].

PHEVs are similar to full hybrids except the battery is charged from the grid, much like a BEV. The larger electrical components of a PHEV allow it to go further on electric driving than a full hybrid. But like all of the vehicle types described earlier, PHEVs include internal combustion engines. They can be run in either serial or parallel hybrid mode. A serial PHEV is essentially an extended range electric vehicle (EREV). In this type of vehicle, all of the driving is electrically powered and the ICE is only used as a generator to recharge the battery. With a parallel hybrid PHEV, both the ICE and the electric motor are connected to the wheels and can drive them directly. And herein lies a major problem with PHEVs: It isn't necessary to actually charge them! In fact, a study by the Miles Consultancy found that many drivers of PHEVs had NEVER charged their vehicles with the plug, due to a lack of infrastructure or a lack of convenience [11]. Unfortunately, using a PHEV in this manner results in greater use-phase CO₂ emissions than its FHEV counterpart [12]. Why? Because this is equivalent to running your PHEV as an FHEV with the extra weight of an unused battery pack! This is a lesson in consumer behavior—it is critical to consider driver habits as we develop future mobility systems. We'll return to this important point later in the book.

Note that all of the vehicle types described above are either full ICE or *electrified*. None of them are actually EVs, but far too often PHEVs and BEVs are grouped together under the “electric vehicle” moniker. This is a bad habit we hope to break with this book. When we group the two together, we will refer to them as “plug-ins” or “plug-in vehicles” (this group also includes EREVs).

The last two vehicle types at the top of the ramp in [Figure 6.1](#) are, in fact, electric vehicles. They feature 100% electric driving with no ICE on board. These include BEVs and FCEVs, which differ in how the electricity is produced. BEVs are charged from the grid while FCEVs use a hydrogen fuel cell to produce electricity (note how we have the FCEV line avoiding the “plug-ins” bar in [Figure 6.1](#)). While both of these vehicle types sound ultra-clean, our colleague Gautam Kalghatgi refers to them as EEVs: emissions elsewhere vehicles. Can you guess why? Spoiler alert: a lot of fossil fuels are involved!

Anatomy of an Electric Vehicle

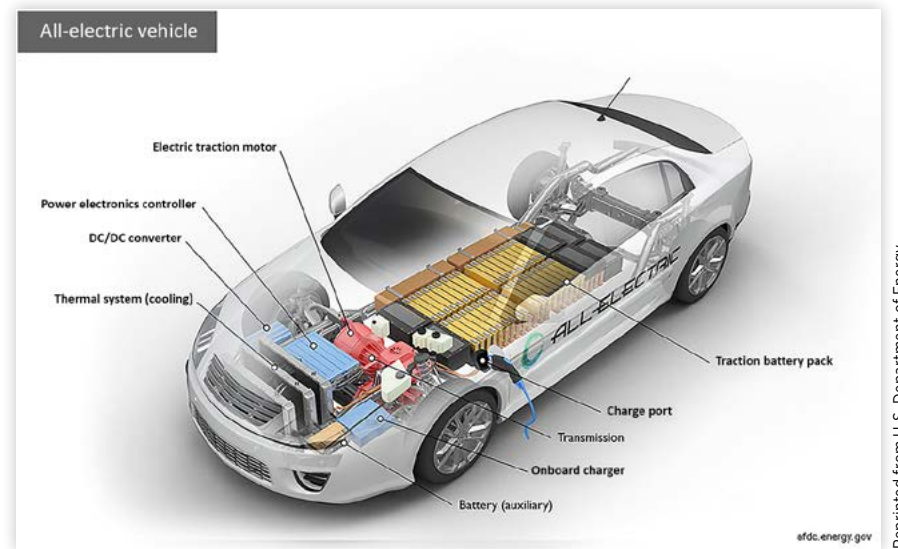
Now that we know the difference between electric and electrified vehicles, let's dive in a bit deeper. How does an EV actually work? We've learned that there are two types of EVs: battery electric and fuel-cell electric. While it's not the goal of this book to go into the gory details of how these vehicles function, we do want the reader to walk away with some basic knowledge. After all, your next car may be electric!

In this chapter, we'll focus on the BEV—but don't worry, we'll get to FCEVs later. The anatomy of a BEV is shown in [Figure 6.2](#). In simple terms, the traction battery pack is charged via the charge port, which is connected to an external power supply during charging. This could happen at home or at a public charging station. Once charged, the electric traction motor uses the electricity stored in the battery pack to generate mechanical power while driving. This power is transferred through the transmission to drive the wheels.

Electric motors are the “business end” of a BEV. They usually consist of two main parts: a rotor (which, funnily enough, rotates) and a stator (which stays still). There are two main motor technologies in use today: permanent magnet and inductance. Typically, we don't actually call them electric motors but electric machines, because they are also capable of generating electrical energy during regenerative braking. A permanent magnet machine (found on the Chevrolet Bolt, Chevrolet Volt, and Tesla 3) uses permanent magnets (usually the stator) alongside windings on the rotor to generate an electric field and provide torque. An inductance machine uses windings on both the rotor and stator to generate the electric field and provide torque. “Standard” BEVs may only use one electric motor, connected to the wheels (usually the front wheels) through a transmission system similar to an ICEV but with fewer gears. Higher performance BEVs such as the Tesla Model S use two motors, one at the front and another at the back. Ultrahigh performance BEVs like the Rimac C_Two have four motors, one for each wheel.

Batteries in BEVs are a little different than those found in ICEVs. The batteries used for traction power delivery operate at much higher voltages (360–450 V) than those used in ICEVs (12 V), but they also have a low-voltage auxiliary battery (also 12 V) for so-called hotel loads (lights, infotainment systems, etc.). In a BEV, the battery is the heaviest and most expensive component. Today, lithium-ion (Li-ion) batteries

FIGURE 6.2 Anatomy of a battery electric vehicle.



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are the dominant technology, but there are other potential technologies in the pipeline. Lithium-ion-phosphate (LiFePO_4) is the most common type of Li-ion battery used in BEVs, but lithium polymer and lithium manganese oxide batteries are also used. In general, a Li-ion battery has two metal sheets, for example, copper and aluminum. Between these metal sheets are the two “poles” that carry positive and negative charge—the cathode and the anode, respectively—and between these two poles the electric “reaction” takes place. For this reaction, a reactive metal such as lithium is required.

The biggest cost factor in a Li-ion battery is the composition of the cathode, the positive pole of the battery. These are usually made of nickel, manganese, and cobalt. Similarly, the anode can be made from graphite powder, lithium, electrolytes, and a separator.

It is common to think of a BEV battery as one homogeneous unit, but it is actually made up of hundreds or thousands of individual cells, most no larger than a standard battery in a TV remote control or your smartphone. As an example, a 2018 24 kWh Nissan LEAF has 192 cells, and a Tesla P100D battery pack (100 kWh, used in the Model X) has 8,256 cells.

Once you have energy stored in these cells, you have to be able to use it. Power electronics are a key component of a BEV in transferring the stored energy from your battery to the electric motors. Power electronics is an umbrella term for a series of electrical components, perhaps a rectifier, DC-DC converter, inverter, and filter, which are required to manage the electrical power within the vehicle. These are used to convert between different voltages (110–240 V AC if charging at home; different if charging from a specialized charger) to the DC voltage needed for your battery.² The DC-DC converter is also used for transferring power between the high- and low-voltage power systems and batteries. The inverter is used to control the electric motor speed and torque, converting the DC current from the battery to the three-phase AC usually needed in the motor(s); it can also accomplish the reverse task during regenerative braking.

With so many cells in a BEV battery, battery management is crucial. A battery management system (BMS) will control the voltage and currents seen by each and every cell, the state of charge of the cells, and the cell temperatures and pressures (you don’t want a cell exploding). The BMS effectively measures and controls the “health” of each cell and shares the load across the cells. All of these parameters are very important for battery life management. Without a good BMS, the battery degrades very quickly (limiting the energy available to be stored, which limits the vehicle range). In particular, thermal management and the frequency of using a fast-charger have been shown to be particularly important parameters for battery life preservation. You may have noticed your smartphone getting hot when being charged through a USB port (5 V). Through a 600 V DC fast charger, things can get very hot

² There’s a lot that could be written about different electrical voltages, AC and DC. AC is the form of electricity you probably interact with most. The outlets in your house are AC, for example. The advantages of AC are that generating and transmitting electricity, as well as changing the voltage of the electricity, are relatively easy. In AC electricity, the current changes direction tens of times every second (i.e., it oscillates). AC generally works better for electric motors. DC, on the other hand, is a flow of electricity a bit like the flow of water in a pipe. This is used in batteries and many digital electronics (anything charged by USB is DC, for example). DC is also used for very long-distance electricity transmission.

very quickly! A Nissan LEAF does not have an active thermal management system, whereas Tesla vehicles do. As a result, LEAF batteries have been shown to degrade much faster than Tesla batteries [13].

Proper thermal management of the battery pack is also crucial for preventing thermal runaway³ and subsequent fire (turning a BEV into an undesirable external combustion vehicle?). Electric motors can also be complicated to cool. The electronics and batteries in BEVs are far more sensitive to temperature changes than the mechanical components of conventional ICE powertrains.

The power and range of a BEV are largely functions of the battery pack size. Range has been a major concern in the past and still is for many BEVs on the market, but you can now drive a fairly long distance if you're willing (and able) to pay for it. The longest range we've seen (so far) is from the Lucid Air Grand Touring edition, with a starting price of \$131,500. It has a projected range of 517 miles (according to Lucid's website) but won't be available until the summer of 2021. Its little brother, the Lucid Air, has a projected range of 406 miles and starts at \$69,900. That version won't be available until "early 2022."

Things are moving fast in the electric vehicle world, but of the cars you could purchase when we wrote this, the Tesla Model S Long Range currently wears the distance crown. It has an EPA estimated range of 373 miles and a starting price of about \$80k (or \$70k if you don't pick the upcoming "autosteer on city streets" option—honestly, it's hard to keep up with Tesla's price changes and software choices!). Tesla currently holds the top four spots, with the Long Range Model 3 coming in with the lowest starting price of around \$50k for a 330 mile range. You can drop below the \$40k mark with a Chevrolet Bolt, but now you're down to 259 miles per charge [14]. All of these vehicles include at least a 10-hour charge time at 240 V, which (if you're in the USA) may be the type of outlet your kitchen range or clothes dryer is plugged into. You know how sometimes you forget to charge your phone at night? Not recommended with your BEV if you need to get to work in the morning.

Eagle-eyed readers may notice that the range for the Model S listed by the EPA doesn't match what's currently quoted on Tesla's website. Why? More updates! In June 2020, Tesla announced that they had broken through the 400-mile range barrier with redesigned wheels, lower rolling resistance tires, revised regenerative braking, and weight reductions [15]. It will be interesting to see if/when the EPA updates their numbers. Did we mention that things are moving fast?

As a reference point on range and cost, a 2020 Hyundai Ioniq Blue, an FHEV, achieves nearly 700 miles per fill up [16]. The starting price for this vehicle is under \$24k. The Hyundai is cheaper, fuels up much faster, and goes significantly further than the Chevrolet Bolt BEV or any of the Tesla models. As an even bigger contrast, at least technology-wise, Felix's 2008 VW Golf diesel (which, don't worry, he hasn't owned for a number of years) could get up to 700 miles and always got 650 miles of real-world (well, Felix's) driving on a full tank. This illustrates that although BEVs hold much promise, they are still limited in terms of cost and practicality.

³ Thermal runaway can occur when some form of stress (e.g., mechanical or thermal) causes an increase in temperature in a battery cell, and then the temperature continues to increase on its own due to a number of exothermic reactions. The increase in temperature can be destructive to the battery—components begin to break down and gases are released within the battery pack, which can then ignite.

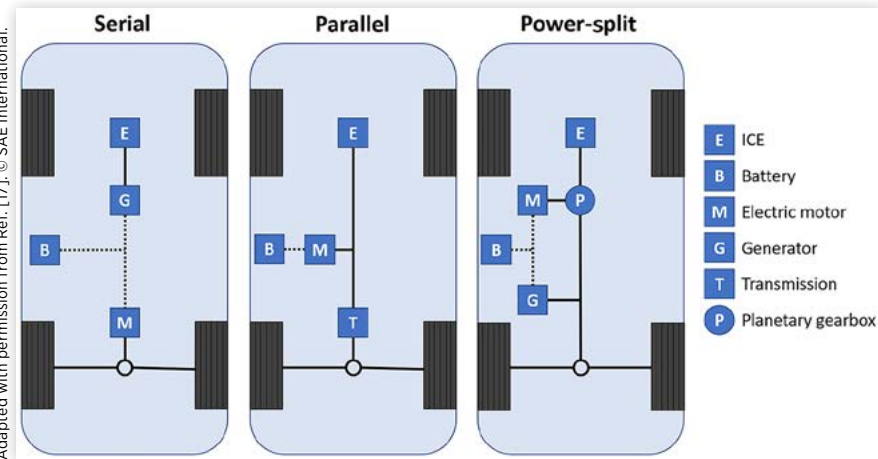
Anatomy of an Electrified (Hybrid) Vehicle

We now know that there are two broad categories for (non-hydrogen) electrified vehicles: battery electric and everything else. The “everything else” is referred to as hybrid vehicles. They are called hybrids because they include both internal combustion engines *and* electric motors. We have already briefly discussed the various types of hybrids in order to untangle the [insert some letter(s) here]EV acronym soup presented earlier in this chapter.

Hybridization offers a number of benefits. First, energy recovery during vehicle braking is straightforward, so much of the fuel energy that is wasted in brakes in an ICEV can be recovered and reused, leading to higher efficiency and lower CO₂ emissions. Some hybrid configurations allow for limited ICE operating points, such that the engine will run where it is most efficient. This again reduces CO₂ emissions. ICEs can be switched off when not needed (e.g., for deceleration and idling). Many hybrid configurations allow for electric-only operation for a limited distance. So, for example, PHEVs could be “geofenced” such that they operate in pure electric mode in low-emissions zones (such as a polluted city center) and then use the ICE outside of these zones. This could quickly improve urban air quality in a way that is feasible in the near future. We’ll discuss how this idea fits into our bigger strategy to go green in Chapter 13.

Hybrid systems can have serial, parallel, or power-split configurations. In serial configurations, electric motors always drive the wheels, and the ICE either charges the battery or directly powers the electric motors through a generator. In this configuration, the engine can run at a predetermined speed where it is most efficient. On the other hand, in parallel configurations, either the electric motors or the ICE can drive the wheels. Typically, the ICE powers the wheels at higher speeds where it is most efficient. The power-split system, which is used in the Toyota Prius, for example, combines the parallel and series configurations through a planetary gearbox. Combining configurations allows for maximum efficiency, power, and driving comfort. [Figure 6.3](#) compares the schematics of these three types of hybrid systems.

FIGURE 6.3 Schematics of serial, parallel, and power-split hybrid systems [17].



History of the Battery Electric Vehicle

We could have named this chapter the “Rise and Fall and Rise and Fall and Rise(?) of the Battery Electric Vehicle,” but is the third rise the charm? Many believe so, and we will discuss the current rise throughout the rest of this book. But we can learn from history, and looking at what worked and what didn’t can help us avoid the pitfalls we encountered in the past.

The Early Days: The First Rise and Fall

As we learned in Chapter 2, BEVs are not new; they have been around since the 19th century! We followed the first rise and fall of the battery electric car, which peaked in 1912, but was virtually driven to extinction some years later by the ICEV. While the early BEVs were quieter, easier to operate, and produced less local pollution than early ICEVs, they couldn’t compete with the longer range, lower cost, and convenience of quickly filling up a gas tank. By 1935, electric cars had essentially disappeared from the roads, and combustion was king.

A Brief Return

This would not be the end of the 20th century BEV. The 1950s and 1960s saw a smattering of experimental BEVs. The Henney Kilowatt (such a great name for an EV!) was introduced in the USA in 1959. The 1960 version had a range of 60 miles and could achieve a speed of 60 mph (sense a theme?), which was quite fast for a BEV at the time [18]. For example, the Battronic Truck Company produced electric trucks in the 1960s that could only reach 25 mph [19]. Shockingly, neither of these vehicles caught on.

FIGURE 6.4 1960 Henney Kilowatt battery electric vehicle.



Courtesy of D. Roberson

FIGURE 6.5 1975 Sebring-Vanguard CitiCar (top) and 1975 Zagato Zele, sold in the USA as the Elcar (bottom).



When the energy crisis of the 1970s hit, some thought the BEV would make an electrifying (ha!) comeback. The US Congress even passed the Electric and Hybrid Vehicle Research, Development and Demonstration Act in 1976 to support R&D of electrified vehicles [20].

Indeed, several electric cars were produced in the 1970s, including the Sebring-Vanguard CitiCar, which sold more than 2,000 units, reached a top speed of 44 mph, and had a range of 50–60 miles [19]. Another 1970s BEV was the Zagato Zele, an Italian car that was sold in the USA as the “Elcar.” The Elcar’s speed and range were similar to the CitiCar, but it fell short in sales [21]. GM also waded into the 1970s BEV craze, debuting their experimental urban electric car at the First Symposium on Low Pollution Power Systems Development in 1973 [22].

From Figures 6.5 and 6.6, you can see that these 1970s cars had a rather unique aesthetic, looking a bit like funky predecessors to the modern Smart car.

With these rather avant-garde options available, why didn’t BEVs take off in the 1970s? Once again, their limited performance and limited range were their downfall. Now, we aren’t considering these experimental BEVs of the 1960s and 1970s as a separate rise and fall since it was more of a blip, but we included it for historical context (and because we had to find an excuse to sneak in those sweet 1970s car pictures!).

FIGURE 6.6 A 1973 GM Urban Electric Car getting a charge in the parking lot at the *First Symposium on Low Pollution Power Systems* in Ann Arbor, Michigan.



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Who Killed the EV1?: The Second Rise and Fall

While the BEV blip of the 1970s was primarily driven by high gasoline costs, in the “modern era” of driving,⁴ renewed interest was brought on by tightening emission standards. But contrary to popular belief, Tesla did not make the first contemporary EV—credit for that goes to General Motors. Their EV1 entered the market in the mid-1990s and was the first mass-produced BEV by a major auto manufacturer in decades.

⁴ Kelly likes to think that the modern era of driving began when he learned how to drive. Please forgive him.

FIGURE 6.7 General Motors EV1.

Approximately 1,100 EV1 cars were produced between 1996 and 1999. However, GM determined they weren't commercially viable and stopped the program after investing more than one billion dollars in production, infrastructure, and marketing. The leased cars were recalled and destroyed in 2003, but some of the technology lives on in GM's hybrid and fuel-cell vehicles [23].

The 2006 documentary/murder mystery *Who Killed the Electric Car?* [24] tries to uncover the answer to its title's question by examining the various "suspects": consumers, the battery technology, oil companies, car companies, the US Government, CARB, and hydrogen fuel cells. According to the film, consumers were not interested, the oil companies felt threatened, car companies were commercially motivated (although, isn't that what companies are supposed to be—commercially motivated?), and so on. One of our favorite quotes from the film was given by Alan Lloyd, CARB Chairman during this period. He said, "Is it our job to clean the air? Or is it to force a certain number of a type of technology on the road?" We would argue strongly for the former, and we'll lay out our case for this in the following chapters.

In an interesting twist, the only suspect that was determined "not guilty" by the film was the technology itself: the batteries. We'll take a look at the evolution of electric car batteries later and decide for ourselves: were they guilty or not?

Prius

Parallel to the EV1, Toyota introduced the world to the Prius, the first mass-produced full hybrid vehicle. As of January 2017, global sales of the Prius family had surpassed

six million units [25], making it the most successful hybrid of all time. While the Prius is an *electrified* vehicle, not an *electric* vehicle, its success still pushed the notion of electric driving further than it had been since 1912. Elon and company may get most of the credit for changing the mobility landscape today, but without Prius there may have never been a Tesla.

Tesla: The Third Rise (with No Fall in Sight)

The destruction of the EV1 fleet motivated the founding of a new company, Tesla Motors (now Tesla, Inc.), by Martin Eberhard and Marc Tarpennin in 2003 [26]. You read that right—although Elon Musk is credited as a co-founder of the company, he was not there at the very beginning. How did he end up with Eberhard and Tarpennin? The story goes that he test drove a tzero, a handmade electric sports car from the company AC Propulsion, of which only three were made. Initially, the tzero was a BEV with a lead-acid battery, but it did come with the option of a trailer carrying an ICE to charge the battery during long-distance travel (that's one way to make an EREV!). The tzero intrigued Musk and he joined Tesla in 2004 with the rights to market the car. The tzero would go on to inspire Tesla's first fully electric vehicle, the Roadster, launched in 2008 [27]. Tesla went on to release the Model S, the Model 3 (Figure 6.8), the Model X, and the Model Y. The Cybertruck and the (new) Roadster are expected to round out the lineup. In 2018, after ten years in the market, Tesla became the best-selling plug-in vehicle manufacturer in the world, with over 245,000 units sold worldwide [28]. In a Twitter post on March 9, 2020, Elon Musk congratulated his team for producing the one-millionth Tesla vehicle, a red Model Y [29].

Tesla has transformed the automotive industry by demonstrating that a car can be both electric and exciting at the same time. It's had its share of ups and downs—and its share of lawsuits and controversies. But as of November 13, 2020, Tesla enjoys an astronomical market cap of over \$380 billion. How high can it go?

FIGURE 6.8 Tesla Model 3.



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Tesla is also now turning a profit—to the tune of \$331 million in the third quarter of 2020. That was Tesla's fifth profitable quarter in a row. But it's important to understand the source of these profits. They are largely due to sales of zero emissions vehicle credits to other car makers, allowing them to meet fuel economy and emissions regulations. So, we have a company whose cars are *not* zero emissions (remember, it's not just about the tailpipe!) making a lot of money selling zero emissions credits to companies who don't yet make zero emissions vehicles. Does anyone else see the irony here?

Regardless, they are currently a profitable company. Will Tesla's bubble burst? It's impossible to predict. What does seem very likely, however, is that Tesla, and the new wave of electric vehicles it has inspired, will be around for the long haul.

What's Different This Time?

Why hasn't Tesla gone the way of the Fritchle or the EV1? Virtually every car manufacturer now sells battery electric cars or plans to begin offering them soon. What's changed?

Well, for one, there's been a renewed focus on the environment and climate change. Prior to the COVID-19 pandemic, the climate crisis was at the forefront of many people's minds. Countries around the globe are focused on reducing reliance on GHG-emitting fossil fuels, and in the transportation sector, many believe this necessitates a transition to electric cars.

The second difference is the implementation of ever-tightening fuel economy/CO₂ regulations, which we discussed in Chapter 4. These stringent fleet-averaged fuel economy requirements (which, if you recall, only take into account tailpipe CO₂) put pressure on the automotive industry to electrify their lineup to remain compliant. After all, BEVs count for 0 g/km CO₂, and in some places several times 0. This allows manufacturers to sell larger vehicles/trucks/SUVs that consumers fervidly want while still meeting fleet-averaged targets.

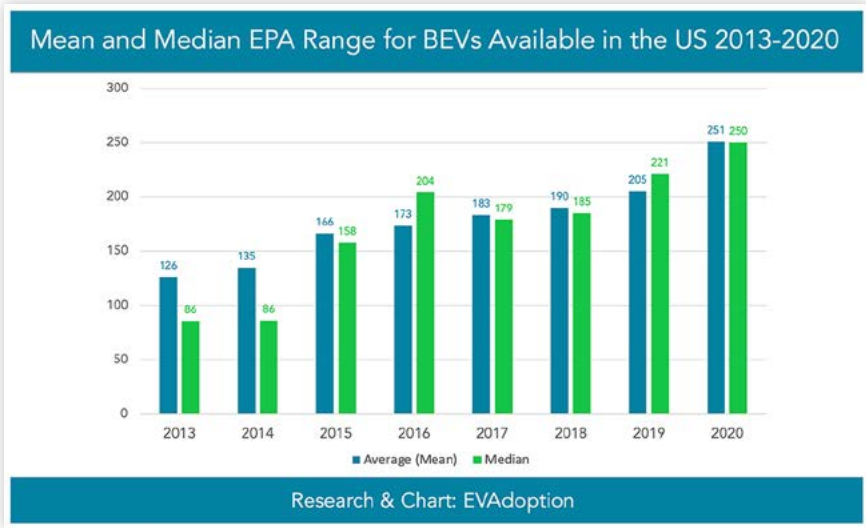
The third difference is the threat of bans on diesel and gasoline cars. Many countries are planning to implement bans on ICEVs and even hybrids at some future date. There is no doubt that this has an impact on what consumers consider purchasing as well as what manufacturers decide to produce.

The fourth difference is subsidies. Many governments are promoting the uptake of BEVs by offering generous subsidies. We'll look further at both the proposed bans and the subsidies in Chapter 12.

And last but not least, the consumer experience is much better than it used to be. Electric cars today are a far cry from the CitiCar or even the EV1, and they keep getting better. Faster charging times, longer range, improved infrastructure—there is still a ways to go to compete with the convenience of a conventional vehicle (or full hybrid, for that matter), but BEVs are definitely headed in the right direction.

In just under a decade, the median range of BEVs in the USA has skyrocketed from 86 miles in 2013 to 250 miles in 2020 (shown in [Figure 6.9](#)). The range is only expected to increase in the future. This longer range, while still lower than many would like, has made the BEV a viable option for a growing portion of the population.

FIGURE 6.9 Mean and median range of BEVs available in the USA from 2013 to 2020 [30].



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Related to experience is cost. While still significantly pricier than their ICEV counterparts, the cost of BEVs continues to come down, as shown in [Figure 6.10](#). Both the improved experience and reduced cost are largely a function of battery technology. Remember that battery technology was the only suspect deemed “not guilty” in the documentary that explores the EV1’s death. However, the battery tech then was not very sophisticated compared to today’s technology. The EV1, initially released in 1996, had a lead-acid battery that provided a range of about 70–90 miles, depending on battery manufacturer [31]. Nickel metal hydride (NiMH) batteries raised the EV1’s range to 160 miles. This was a major improvement, but with an eight-hour charging time, was it enough to keep this car alive against the long range and fueling convenience of a conventional vehicle? We would argue that the technology was just as “guilty” as the other suspects, but perhaps the jury is still out on that one ...

As we saw earlier, today’s vehicle battery technology is based on lithium-ion batteries, originally developed for consumer electronics. You can see why they’re appealing in [Figure 6.11](#)—compared to lead-acid and NiMH, Li-ion batteries have a much higher energy density, allowing for smaller and lighter batteries. Note that the prices shown in [Figure 6.10](#) are for Li-ion battery packs. Some believe that we’ll need to reach \$100/kWh before BEVs can really compete with conventional vehicles on a large scale [32]. This is projected to occur in 2024 by BloombergNEF, as shown in [Figure 6.10](#).

Of course, new battery chemistries and cell technologies are the focus of many startups, much venture capital, and a growing amount of government funding.

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FIGURE 6.10 Average electric car battery price from 2010 to 2019. Also shown is a projected cost in 2024.

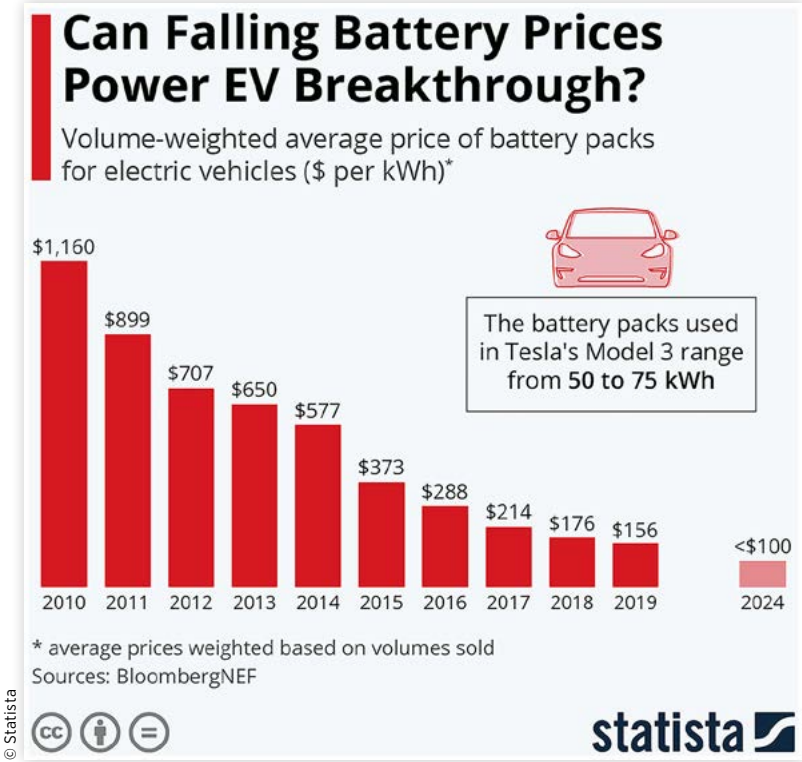
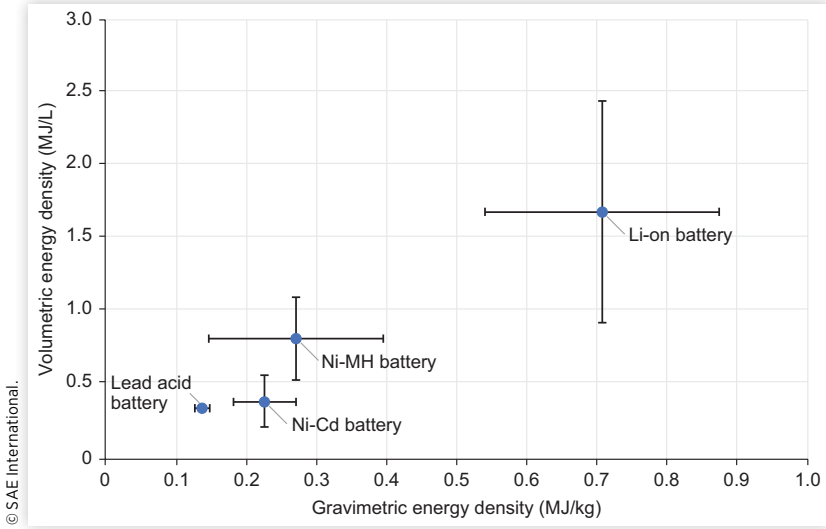


FIGURE 6.11 Comparison of volumetric and gravimetric energy density for various battery types.



For example, the US Department of Energy (DOE) Vehicles Technology Office has the goal to [33]:

... identify new battery chemistry and cell technologies with the potential to reduce the cost of electric vehicle battery packs by more than half, to less than \$100/kWh (ultimate goal is \$60/kWh battery cell cost), increase range to 300 miles, and decrease charge time to 15 minutes or less by 2028.

The funded projects include “next generation lithium-ion battery technologies” (e.g., advanced anodes, such as silicon, advanced cathodes, no-cobalt/low-cobalt cathodes, and graphene), as well as “beyond lithium-ion battery technologies” (e.g., metallic lithium, solid-state, lithium sulfur, lithium-air, and sodium-ion). These technologies are being explored by manufacturers as well, but are all at an early stage of development.

In addition, Tesla (naturally) isn’t stagnant in this area. Elon Musk recently revealed his plans to revolutionize the battery, with the ultimate goal of producing a \$25k vehicle in just three years. The announcement came on “Battery Day,” held after Tesla’s annual stockholders meeting on September 22, 2020. Included in Tesla’s plans is a switch from cobalt to nickel cathodes and a redesign of their battery cells. We’re not holding our breath for that cheap Tesla, but check back with us in September 2023!

The Last Mile

We’ve covered a lot of ground in this chapter. First and foremost, we’ve tried to clear up any confusion about the difference between electric and electrified vehicles. We have also outlined the many propulsion technologies that exist today, from pure combustion to pure electrification and everything in between.

We traced the history of the BEV, from its first rise and fall over 100 years ago up to today. The BEV’s journey has been riddled with speed bumps, and really, the fundamental challenges preventing past adoption still exist today. But the technology keeps improving, and, like it or not, the times they are a-changin’. The combined forces of the Paris Agreement, Dieselgate, and Tesla have brought the BEV to the forefront of the future mobility discussion.

Now that we know BEVs are likely around for the long haul, what are the pros and cons of a modern BEV? How does it compare to its competitors? We will attempt to answer these questions in the next chapter.